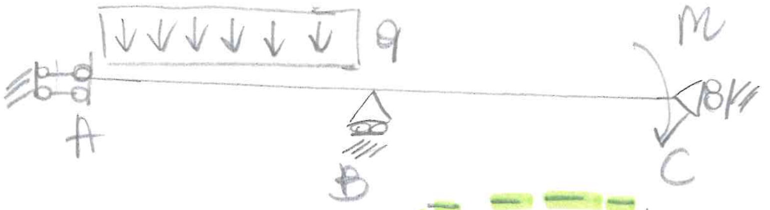
Metodo degli SPOSTAMENTI:

caso 1



$3t - s = l - i$

$3 - 4 = -1$

$l - i = -1$ ∄C $i = 1$ → Non Labile 1-volta iperst.

L'eq.ne della linea elastica $\boxed{EIv^{IV} = q}$ nasce, combinando l'equilibrio locale, legame costitutivo + congruenza

non necessita che venga risolto il problema dell'equilibrio in forma chiusa.

MEMO:

$N = EAw'$

$M = -EIv''$

$T = -EIv'''$

$\varphi = -v'$

$\underbrace{EIv^{IV} = q}$ vale sempre indip. dai vincoli → Eq.ne di campo

descrive tutte le possibili deformate compatibili con la fisica locale in funzione dei vincoli si estrae quelle reali → l'iperstaticità interviene nella determ. delle costanti (BCs)

1 corpo → 2 tratti matematici AB - BC

AB

$$\begin{cases} EAw'' = 0 \\ EIv^{IV} = q \end{cases} \longrightarrow \quad EAw' = C_1 \rightarrow EAw = C_1 z + C_2$$

$$\rightarrow EIv''' = qz + C_3 \rightarrow EIv'' = \frac{qz^2}{2} + C_3 z + C_4 \rightarrow EIv' = \frac{qz^3}{6} + \frac{C_3 z^2}{2} + C_4 z + C_5$$

$$EIv = \frac{C_3 z^3}{6} + \frac{C_4 z^2}{2} + C_5 z + C_6 + \frac{qz^4}{24}$$

6 costanti

BC

$$\begin{cases} EAw'' = 0 \\ EIv^{IV} = 0 \end{cases} \longrightarrow \quad EAw = C_7 z + C_8$$

$$EIv = \frac{C_9 z^3}{6} + \frac{C_{10} z^2}{2} + C_{11} z + C_{12}$$

6 costanti

1 iperst → #BCs < #BCC

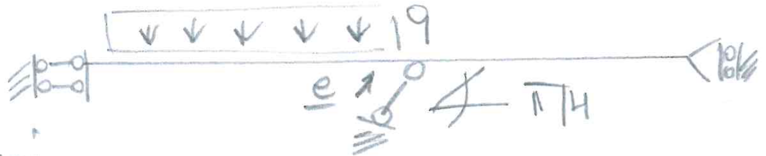
BCC

A) $w_A = 0$, $\varphi_A = 0$ • B) $\Delta w_B = 0$, $\Delta v_B = 0$, $\Delta\varphi_B = 0$, $v_B^+ = 0$ • C) $w_C = 0$ (7)

BCS

A) $T_A = 0$ • B) $\Delta N_B = 0$, $\Delta M_B = 0$ • C) $M_C = -M$, $T_C = 0$ (5)

caso 2



$\underline{e} = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

Eq^{ni} Linea Elastica di IV ordine uguali a prima cambiano le BC in B

- $BCC_{(B)}$: $\underline{s}_B^+ \cdot \underline{e} \overset{!}{=} 0$ $\quad w_B^+ \frac{\sqrt{2}}{2} + v_B^+ \frac{\sqrt{2}}{2} = 0$ $\quad \boxed{w_B^+ = -v_B^+}$

  $\Delta w_B = 0$

  $\Delta v_B = 0$

  $\Delta \varphi_B = 0$

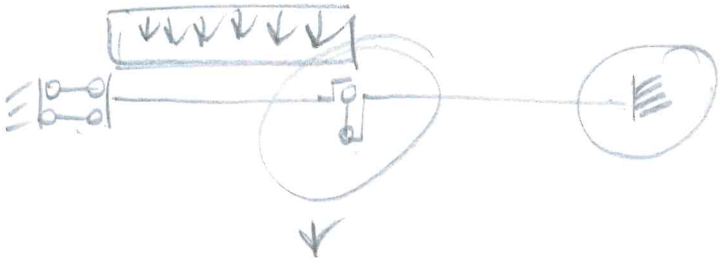
$\underline{e}^\perp = \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

- $BCS_{(B)}$ $\quad \Delta M_B = 0$

  $\begin{Bmatrix} \Delta N_B \\ \Delta T_B \end{Bmatrix} \cdot \underline{e}^\perp = 0 \qquad (N_B^+ - N_B^-)\frac{\sqrt{2}}{2} - (T_B^+ - T_B^-)\frac{\sqrt{2}}{2} \overset{!}{=} 0$

  nella direzione $\perp$ al pendolo siamo certi che il pendolo non esplica reazioni

caso 3



→ solo cinem.

$BCC_{(B)}$ $\Delta v_B = 0$

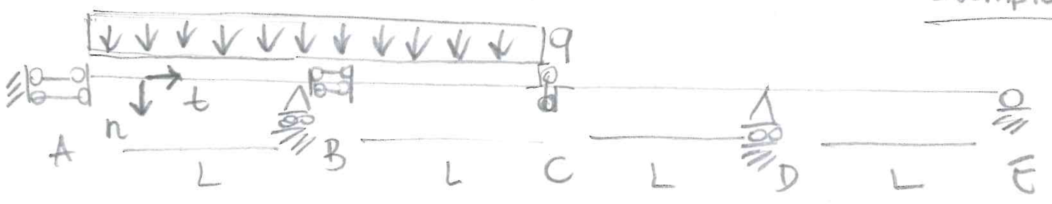
BCS $\quad \Delta N_B = 0 \quad N_B^+ = 0$

$\Delta T_B = 0$

$\Delta M_B = 0 \quad M_B^+ = 0$

non saltano e sono nulle

METODO DEGLI SPOSTAMENTI (Linea Elastica IV ordine) Esempio 1



$3t - s = \ell - i$ $\quad 9-9=0 \quad \ell=1 \quad \not{C} \quad \ell=0 \quad i=0 \rightarrow$ Isostatica #BCC = #BCS

$t=3$ 4 tratti matematici

$EA\,w''(z) = -p$

$EI\,v^{IV}(z) = q$

AB $z \in (0,L)$

$EA\,w''(z) = 0 \rightarrow EA\,w'(z) = C_1 \rightarrow EA\,w(z) = C_1 z + C_2$

$EI\,v^{IV}(z) = q \rightarrow EI\,v'''(z) = qz + C_3 \rightarrow EI\,v'' = q\frac{z^2}{2} + C_3 z + C_4$

$EI\,v'(z) = \frac{qz^3}{6} + C_3\frac{z^2}{2} + C_4 z + C_5$

$EI\,v(z) = \frac{qz^4}{24} + C_3\frac{z^3}{6} + C_4\frac{z^2}{2} + C_5 z + C_6$

BC $z \in (0,L)$

$EA\,w''(z) = 0 \rightarrow EA\,w'(z) = C_7 \rightarrow EA\,w(z) = C_7 z + C_8$

$EI\,v^{IV}(z) = q \rightarrow EI\,v'''(z) = qz + C_9 \rightarrow EI\,v''(z) = q\frac{z^2}{2} + C_9 z + C_{10}$

$EI\,v''(z) = q\frac{z^3}{6} + C_9\frac{z^2}{2} + C_{10} z + C_{11}$

$EI\,v(z) = q\frac{z^4}{24} + C_9\frac{z^3}{6} + C_{10}\frac{z^2}{2} + C_{11} z + C_{12}$

CD $z \in (0,L)$

$EA\,w'(z) = 0 \rightarrow EA\,w'(z) = C_{13} \rightarrow EA\,w(z) = C_{13} z + C_{14}$

$EI\,v^{IV}(z) = 0 \rightarrow EI\,v'''(z) = C_{15} \rightarrow EI\,v''(z) = C_{15} z + C_{16}$

$EI\,v'(z) = C_{15}\frac{z^2}{2} + C_{16} z + C_{17}$

$EI\,v(z) = C_{15}\frac{z^3}{6} + C_{16}\frac{z^2}{2} + C_{17} z + C_{18}$

DE $z \in (0,L)$

$EA\,w''(z) = 0 \rightarrow EA\,w'(z) = C_{19} \rightarrow EA\,w(z) = C_{19} z + C_{20}$

$EI\,v^{IV}(z) = 0 \rightarrow EI\,v'''(z) = C_{21} \rightarrow EI\,v''(z) = C_{21} z + C_{22}$

$EI\,v'(z) = C_{21}\frac{z^2}{2} + C_{22} z + C_{23}$

$EI\,v(z) = C_{21}\frac{z^3}{6} + C_{22}\frac{z^2}{2} + C_{23} z + C_{24}$

BCC

A) $w_A = 0$ $\varphi_A = 0$

B) $\Delta w_B = 0$ $(w_B^+ - w_B^- = 0)$
$\Delta \varphi_B = 0$ $(\varphi_B^+ - \varphi_B^- = 0)$
$v_B^- = 0$

12

C) $\Delta v_C = 0$ $(v_C^+ - v_C^- = 0)$

D) $\Delta w_D = 0$ $(w_D^+ - w_D^- = 0)$
$\Delta \varphi_D = 0$ $(\varphi_D^+ - \varphi_D^- = 0)$
$\Delta v_D = 0$ $(v_D^+ - v_D^- = 0)$
$v_D^+ = 0$

E) $v_E = 0$ $w_E = 0$

BCS

A) $T_A = 0$

B) $\Delta M_B = 0$ $(M_B^+ - M_B^- = 0)$
$\Delta N_B = 0$ $(N_B^+ - N_B^- = 0)$
$T_B^+ = 0$

C) $\Delta T_C = 0$ $(T_C^+ - T_C^- = 0)$
$\Delta M_C = 0$ $M_C^+ = 0$
$\Delta N_C = 0$ $N_C^+ = 0$ *

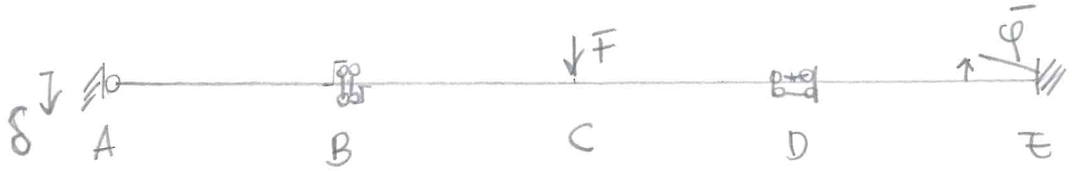
* non saltano e devono essere zero
il carrello non esplica quelle reazioni

D) $\Delta M_D = 0$ $(M_D^+ - M_D^- = 0)$
$\Delta N_c = 0$ $(N_c^+ - N_c^- = 0)$

E) $M_E = 0$

12

Esempio 2



$3t - S = l - 1$

$9 - 9 = l - 1$

$\not{X}C \; l = 0$

$\lambda = 0$

$EA w''(z) = -p$

$EI v^{IV}(z) = q$

AB

$EA w''(z) = 0 \rightarrow EA w'(z) = C_1 \rightarrow EA w(z) = C_1 z + C_2$

$EI v^{IV}(z) = 0 \rightarrow EI v'''(z) = C_3 \rightarrow EI v''(z) = C_3 z + C_4 \rightarrow EI v'(z) = C_3 \frac{z^2}{2} + C_4 z + C_5$

$EI v(z) = C_3 \frac{z^3}{6} + C_4 \frac{z^2}{2} + C_5 z + C_6$

BC

$EA w''(z) = 0 \; \text{-----} \rightarrow EA w(z) = C_7 z + C_8$

$EI v^{IV}(z) = 0 \; \text{---} \rightarrow EI v(z) = C_9 \frac{z^3}{6} + C_{10} \frac{z^2}{2} + C_{11} z + C_{12}$

CD

$EA w''(z) = 0 \; \text{----} \rightarrow EA w(z) = C_{13} z + C_{14}$

24 costanti

$EI v'(z) = 0 \; \text{---} \rightarrow EI v(z) = C_{15} \frac{z^3}{6} + C_{16} \frac{z^2}{2} + C_{17} z + C_{18}$

DE

$EA w''(z) = 0 \; \text{---} \rightarrow EA w(z) = C_{19} z + C_{20}$

$EI v^{IV}(z) = 0 \; \text{---} \rightarrow EI v(z) = C_{21} \frac{z^3}{6} + C_{22} \frac{z^2}{2} + C_{23} z + C_{24}$

BCC = # BCS

BCC

A) $w_A = 0$
$v_A = \delta$

B) $\Delta v_B = 0 \quad (v_B^+ - v_B^- = 0)$
$\Delta \varphi_B = 0 \quad (\varphi_B^+ - \varphi_B^- = 0)$

C) $\Delta w_C = 0 \quad (w_C^+ - w_C^- = 0)$
$\Delta v_C = 0 \quad (v_C^+ - v_C^- = 0)$
$\Delta \varphi_C = 0 \quad (\varphi_C^+ - \varphi_C^- = 0)$

D) $\Delta w_D = 0 \quad (w_D^+ - w_D^- = 0)$
$\Delta \varphi_D = 0 \quad (\varphi_D^+ - \varphi_D^- = 0)$

E) $w_E = 0$
$v_E = 0$
$\varphi_E = -\bar{\varphi}$

12

BCS

A) $M_A = 0$

B) $\Delta T_B = 0$
$\Delta M_B = 0$
$\Delta N_B = 0 \quad N_B^+ = 0$

C) $\Delta N_C = 0$
$\Delta M_C = 0$
$\Delta T_C = -F$

D) $\Delta M_D = 0$
$\Delta N_D = 0$
$\Delta T_D = 0$
$T_D^+ = 0$

12

BCS

A) $M_A = 0$

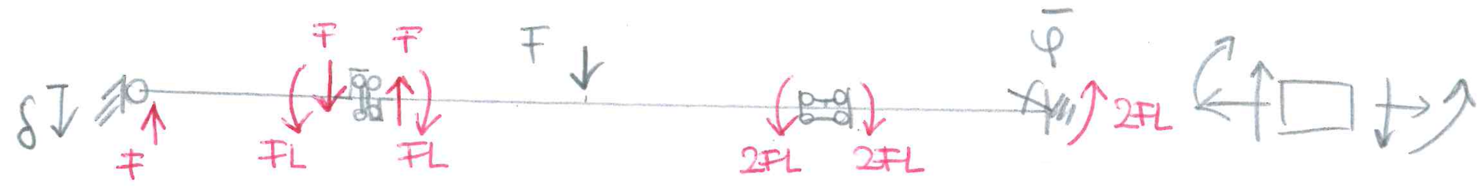
B) $\Delta T_B = 0 \quad (T_B^+ - T_B^- = 0)$
$\Delta M_B = 0 \quad (M_B^+ - M_B^- = 0)$
$\Delta N_B = 0 \quad + N_B^+ = 0$

C) $\Delta M_C = 0 \quad (M_C^+ - M_C^- = 0)$
$\Delta N_C = 0 \quad (N_C^+ - N_C^- = 0)$
$\Delta T_C = -F \quad (T_C^+ - T_C^- = -F)$

D) $\Delta M_D = 0 \quad (M_D^+ - M_D^- = 0)$
$\Delta N_D = 0 \quad (N_D^+ - N_D^- = 0)$
$\Delta T_D = 0 \quad + T_D^+ = 0$

12

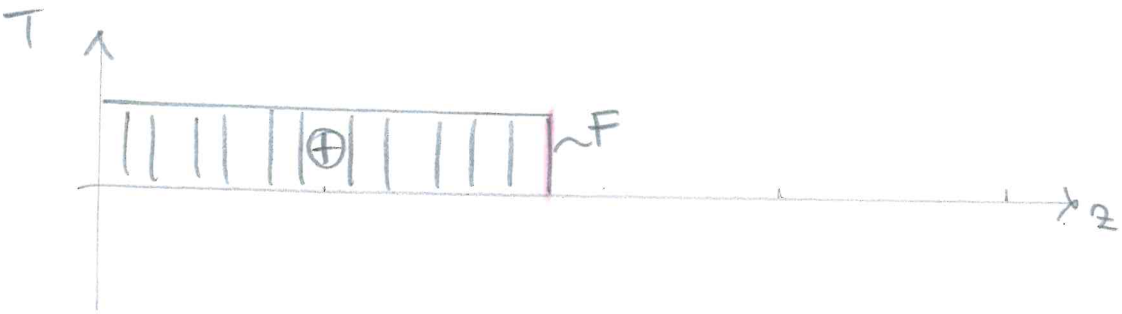
Reazioni vincolari + cds

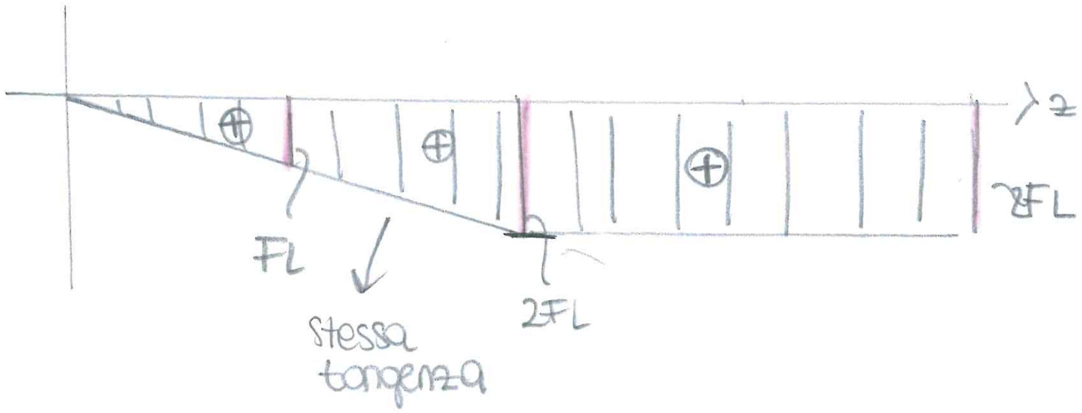


$N(z) = 0 \quad \forall z$

$$T(z) = \begin{cases} F & \text{su } AB \quad z \in (0, L) \\ F & \text{su } BC \quad z \in (0, L) \\ 0 & \text{su } CD \quad z \in (0, L) \\ 0 & \text{su } DE \quad z \in (0, L) \end{cases}$$

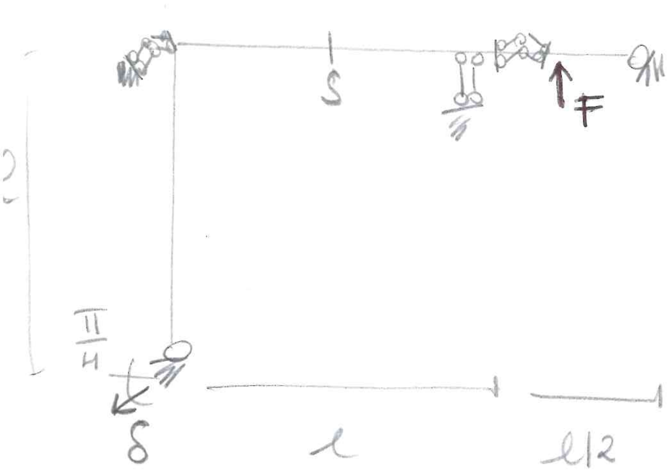
$$M(z) = \begin{cases} F \cdot z & \text{su } AB \\ FL + Fz & \text{su } BC \\ 2FL & \text{su } CD \\ 2FL & \text{su } DE \end{cases}$$





TLV IPERSTATICO!

? vs



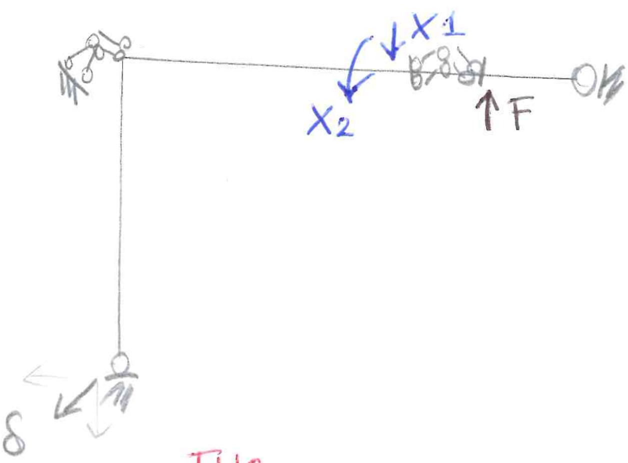
$3t - s = l - i$

$t = 2 \quad s = \Sigma MS = 8$

$6 - 8 = -2 \quad \nexists C \rightarrow l = 0$

$i = 2$

S^{eff}

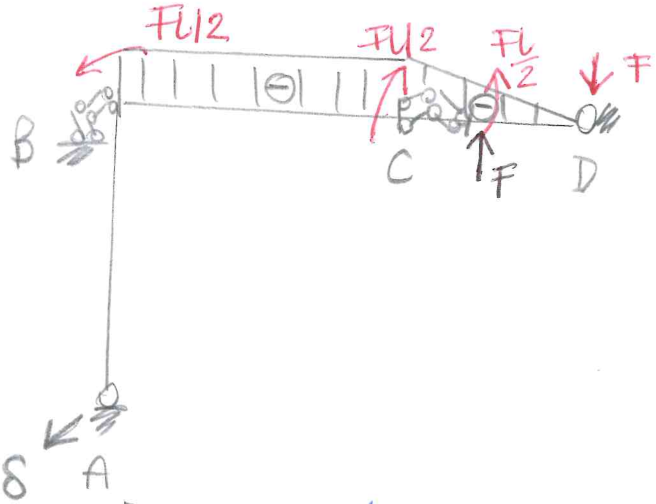


$S^{eff} = S^0 + X_1 S^1 + X_2 S^2$

$+ \quad v_C = 0$

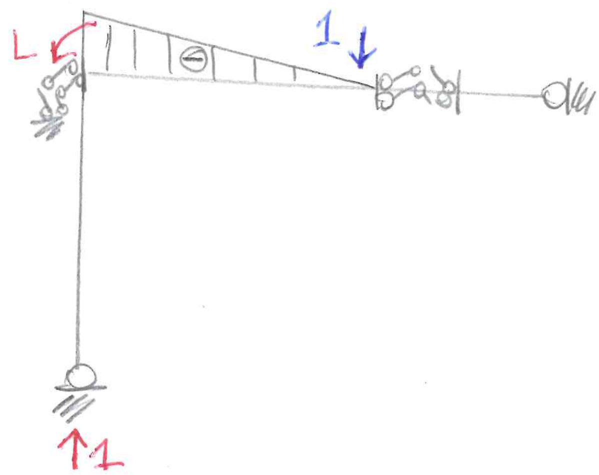
$\varphi_C = 0$

S^0



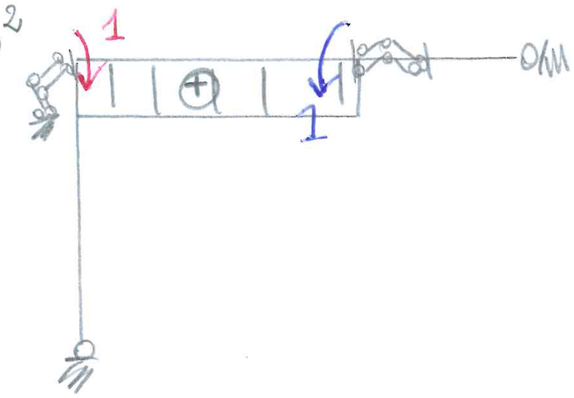
$$\overset{0}{m}(z) = \begin{cases} 0 & \text{su } AB \\ -Fl/2 & \text{su } BC \\ -Fl/2 + F \cdot z & \text{su } CD \end{cases}$$

S^1



$$M^1(z) = \begin{cases} 0 & \text{su } AB \\ -(L - z) & \text{su } BC \\ 0 & \text{su } CD \end{cases}$$

S^2



$$m^2(z) = \begin{cases} 0 & \text{su } AB \\ 1 & \text{su } BC \\ 0 & \text{su } CD \end{cases}$$

$$S^s = S^{eff} = S^0 + X_1 S^1 + X_2 S^2$$
$$S^f = S^1$$

$$L_{ve} = 1 \cdot \delta\sqrt{2}/2 + 1 \cdot \underbrace{v_C^-}_{0} \to \frac{\delta\sqrt{2}}{2}$$

$$L_{vi} = \int m^f \cdot \frac{m^s}{EI} \, dz$$

$$\overset{!}{=} \frac{1}{EI} \int_A^B (L-z) \cdot \left(-\frac{FL}{2} - X_1(L-z) + X_2 \cdot 1\right) dz$$

$$S^s = S^{eff}$$
$$S^f = S^2$$

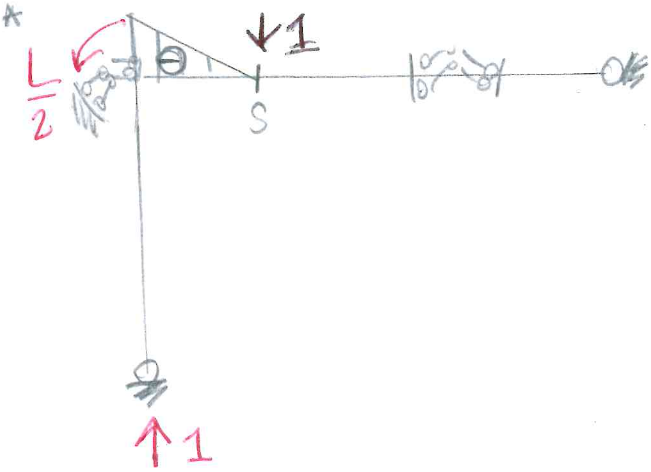
$$L_{ve} = 1 \cdot \varphi_C^- \overset{!}{=} 0$$

$$L_{vi} = \int m^f \cdot \frac{m^s}{EI} \, dz$$

$$\overset{!}{=} \frac{1}{EI} \int_A^B 1 \cdot \left(-\frac{FL}{2} - X_1(L-z) + X_2 \cdot 1\right) dz$$

TLV per il calcolo di VS

S^*



$$m^*(z) = \begin{cases} 0 & \text{su } AB \\ -\frac{L}{2} + z & \text{su } BS \\ 0 & \end{cases}$$

$$S^s = S^{eff}$$
$$S^f = S^*$$

$$L_{ve} = V_S - \delta\sqrt{2}/2$$

$$L_{vi} = \int_B^S m^* \cdot \frac{m^s}{EI} \, dz = \int_0^{L/2} \left(-\frac{L}{2} + z\right)\left(-\frac{FL}{2} - X_1(L-z) + X_2(1)\right) dz$$